

UNIT 1: Atomic & Molecular Structure

1. Molecular Orbital Theory (MOT)

Think of MOT as a partnership: when two individual atoms come together, their atomic orbitals merge completely to form brand new Molecular Orbitals (MOs) that belong to the entire molecule.

Core Postulates of LCAO Principle:

- LCAO Principle: Atomic Orbitals combine linearly by addition ($\psi_A + \psi_B$ Constructive interference) or subtraction ($\psi_A - \psi_B$ Destructive interference).
- Bonding MOs (σ, π): Lower energy, higher stability than isolated atomic orbitals. Electron density accumulates between nuclei.
- Antibonding MOs (σ^*, π^*): Higher energy, lower stability. Electron density drops to zero at the nodal plane between nuclei.

• Bond Order Formula:

$$\text{Bond Order} = \frac{1}{2} (N_b - N_a)$$

where N_b = number of bonding electrons, N_a = number of antibonding electrons.

Homonuclear Diatomic Molecules (N_2 & O_2):

- Nitrogen (N_2 - 14 e^-): Electronic Config: $\sigma 1s^2 \sigma^* 1s^2 \sigma 2s^2 \sigma^* 2s^2 (\pi 2p_x^2 = \pi 2p_y^2) \sigma 2p_z^2$.

$N_b = 10, N_a = 4$ Bond Order = $\frac{1}{2}(10 - 4) = 3$ (Triple Bond). All electrons are paired Diamagnetic.

- Oxygen (O_2 - 16 e^-): Electronic Config: $\sigma 1s^2 \sigma^* 1s^2 \sigma 2s^2 \sigma^* 2s^2 \sigma 2p_z^2 (\pi 2p_x^2 = \pi 2p_y^2) (\pi^* 2p_x^1 = \pi^* 2p_y^1)$.

$N_b = 10, N_a = 6$ Bond Order = $\frac{1}{2}(10 - 6) = 2$ (Double Bond). Contains 2 unpaired electrons in π^* Paramagnetic.

Heteronuclear Diatomics (CO & NO):

- Carbon Monoxide (CO - 14 e^-): O is more electronegative than C. Bond order = 3, highly stable.

- Nitric Oxide (NO - 15 e^-): Has 1 unpaired electron in $\pi^* 2p$ orbital. Bond order = 2.5. Paramagnetic.

2. Band Theory of Solids

In a solid crystal, billions of atomic orbitals overlap to form continuous energy bands. Valence electrons form the Valence Band (VB) and empty upper states form the Conduction Band (CB).

Classification based on Band Gap (E_g):

1. Conductors (Metals): VB and CB overlap directly ($E_g = 0$). Electrons move freely upon applying voltage.
 2. Insulators (Diamond, Wood): Huge forbidden band gap ($E_g > 3$ eV). Thermal energy cannot promote electrons to CB.
 3. Semiconductors (Si, Ge): Small energy gap ($E_g \sim 1$ eV). At 0K acts as insulator; at room temp, thermal excitation promotes electrons.
- Doping: Intrinsic vs Extrinsic. n-type: Doped with Group 15 (P/As) excess e^- . p-type: Doped with Group 13 (B/Ga) excess holes.

Diagram Sketch: [Valence Band] — (small 1eV gap) — [Conduction Band] (Semiconductor).

3. Liquid Crystals & Applications

Liquid crystals exist in a state of matter between crystalline solid and isotropic liquid (called Mesophase). They flow like liquid but possess molecular orientational order like a crystal!

Structural Classification of Liquid Crystals:

- Nematic Phase: Molecules are aligned parallel along a long axis (orientational order) but have no positional layers. Looks thread-like under microscope.
- Smectic Phase: Molecules have orientational order AND form distinct parallel layers. Feels soap-like.
- Cholesteric Phase: Layered structure with each layer rotated slightly, forming a helical spiral (optically active, changes color with temperature!).

Engineering Applications:

1. Liquid Crystal Displays (LCDs): Applied electric field reorients nematic molecules in watches and laptops.
2. Thermography: Cholesteric liquid crystals change color with temperature skin fever sensors & circuit hotspot detection.

4. Quick Exam Revision Summary

- Bond Order = $\frac{1}{2} (N_b - N_a)$. N_2 (BO=3, Diamagnetic) vs O_2 (BO=2, Paramagnetic).
- Semiconductor Gap $E_g \sim 1$ eV; Metal $E_g = 0$.
- Nematic (thread-like) vs Smectic (layered) vs Cholesteric (helical spiral).